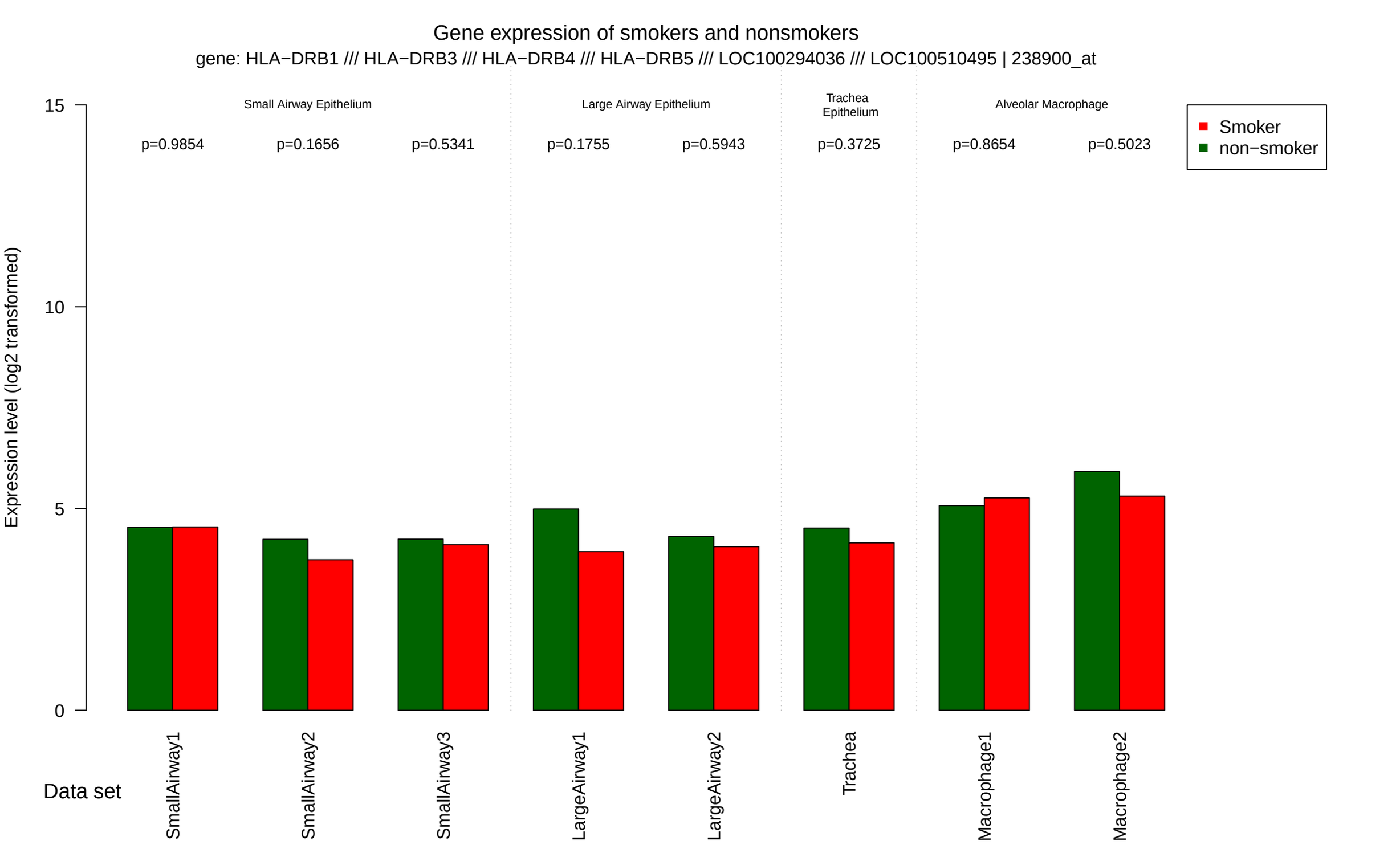
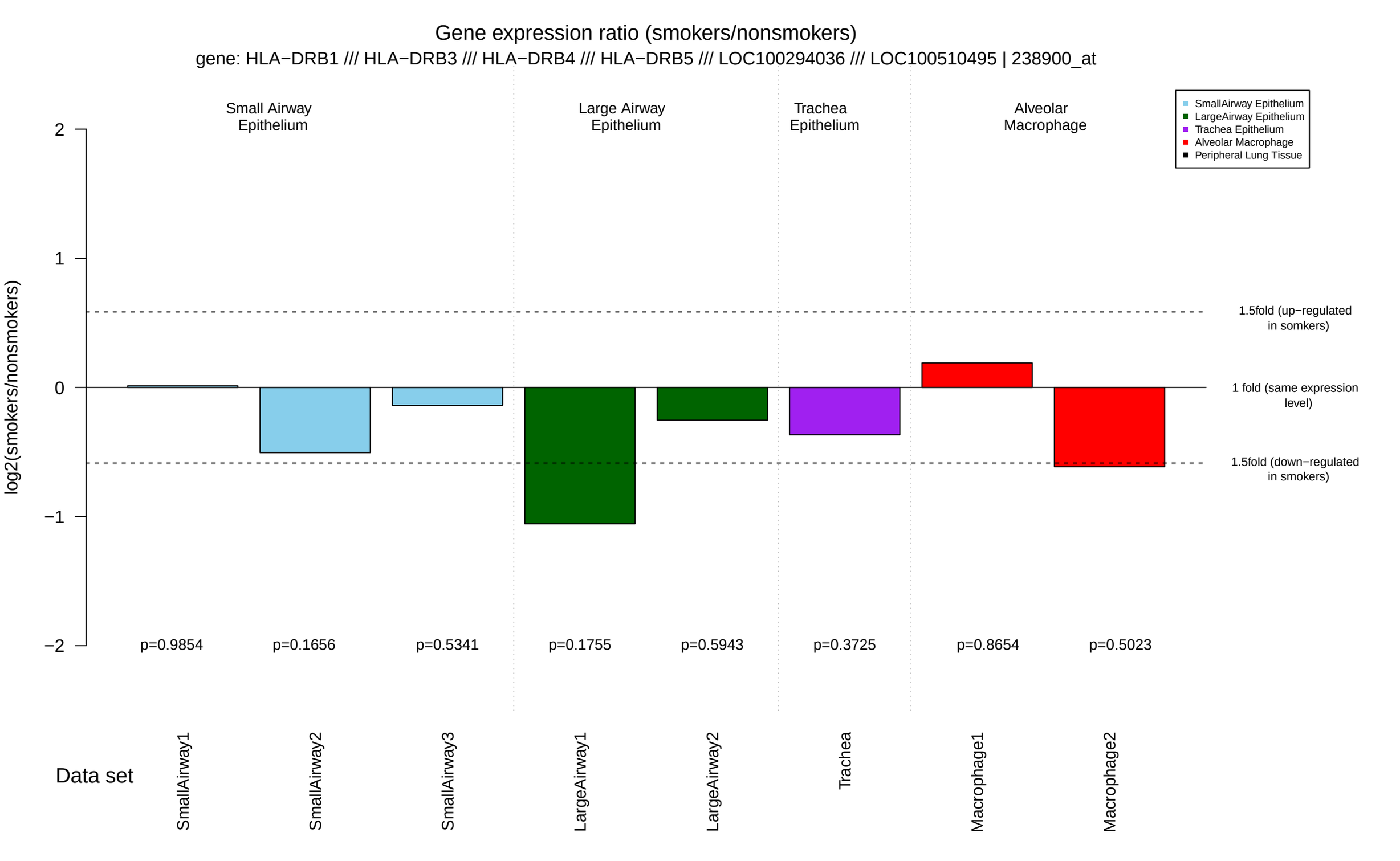
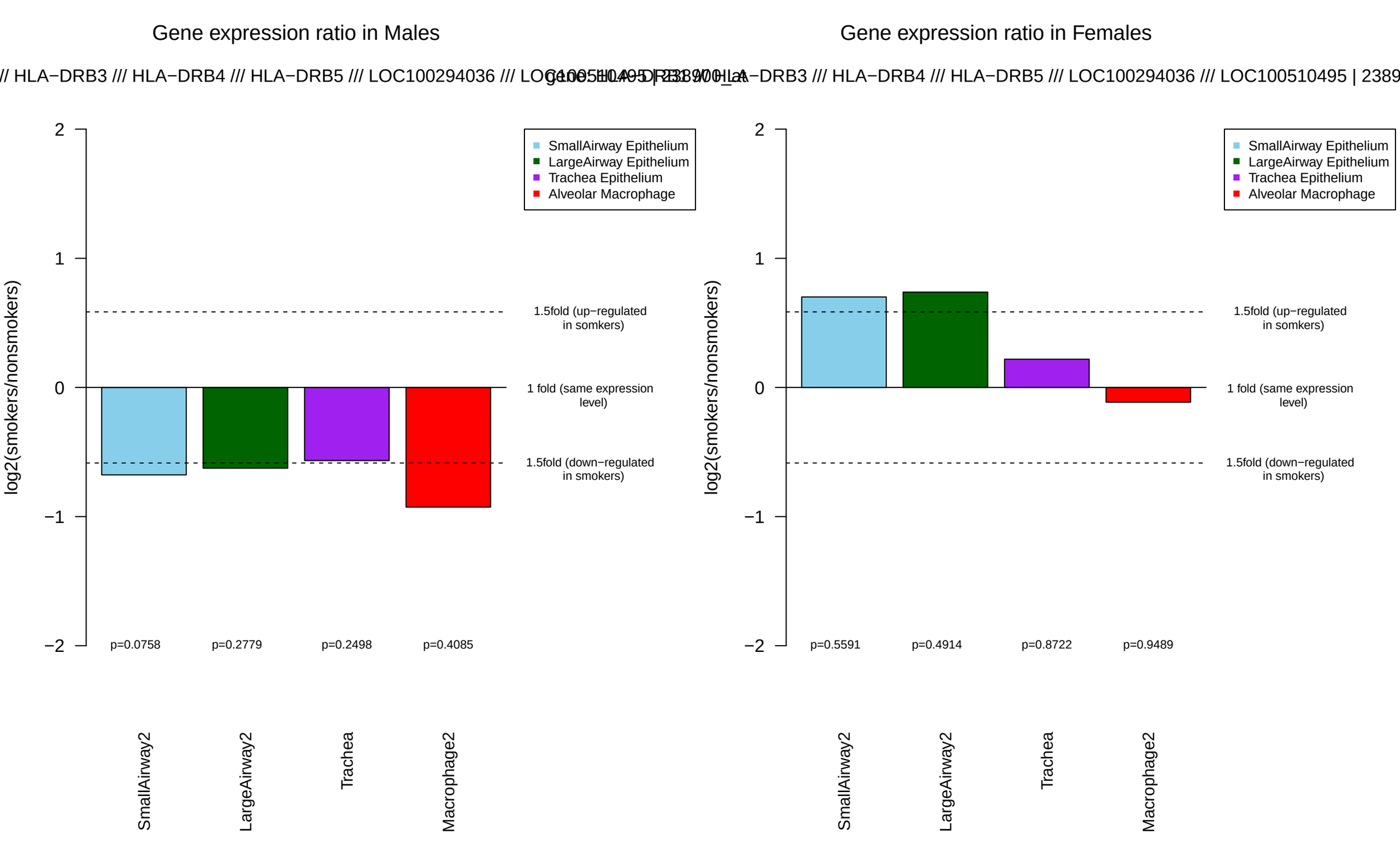
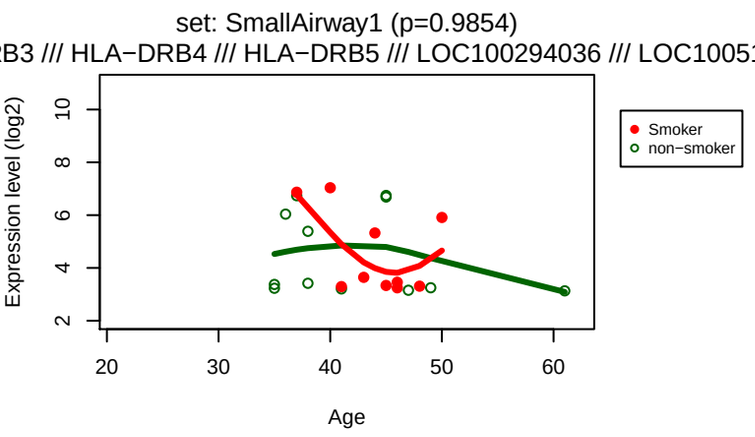
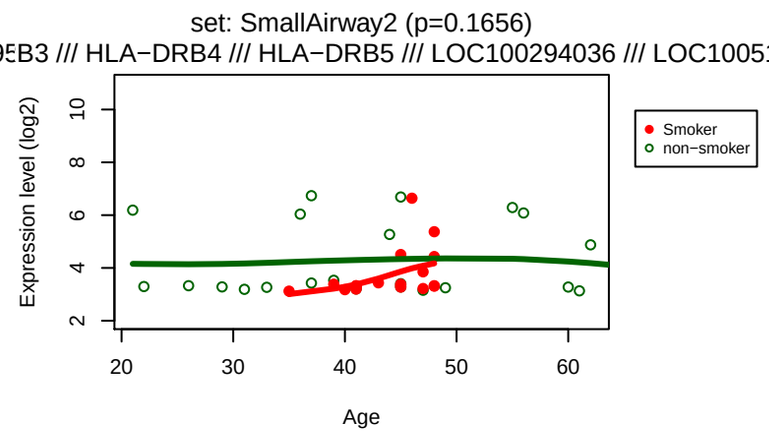
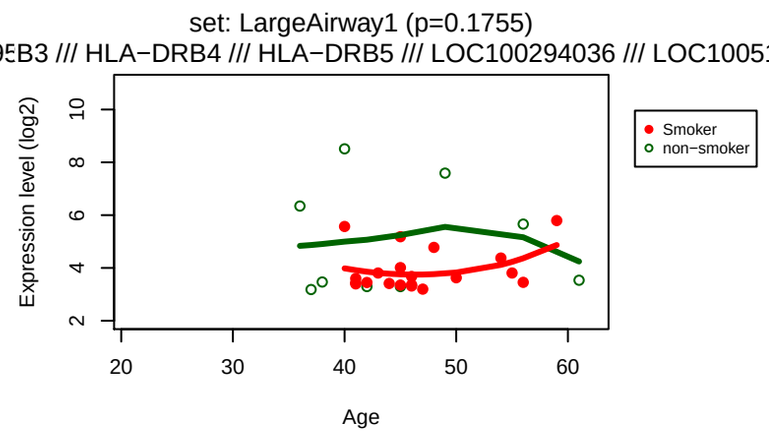

Gene expression of smokers and nonsmokers
gene: HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC100510495 | 238900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9854
p=0.1656
p=0.5341
p=0.1755
p=0.5943
p=0.3725
p=0.8654
p=0.5023
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

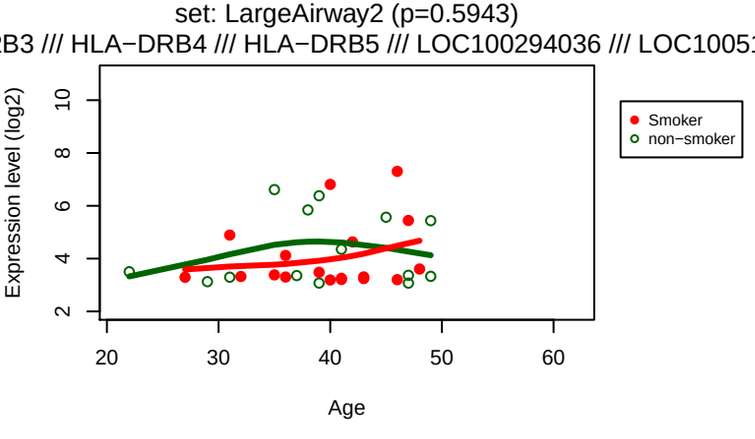

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC100510495 | 238900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9854
p=0.1656
p=0.5341
p=0.1755
p=0.5943
p=0.3725
p=0.8654
p=0.5023
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

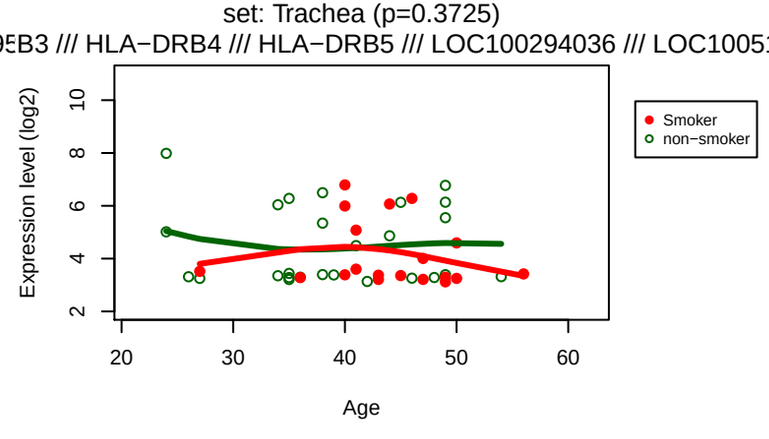

Gene expression ratio in Males
// HLA−DRB3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC100510495 | 23890_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0758
p=0.2779
p=0.2498
p=0.4085
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
// HLA−DRB3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC100510495 | 23890_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5591
p=0.4914
p=0.8722
p=0.9489
SmallAirway2
LargeAirway2
Trachea
Macrophage2

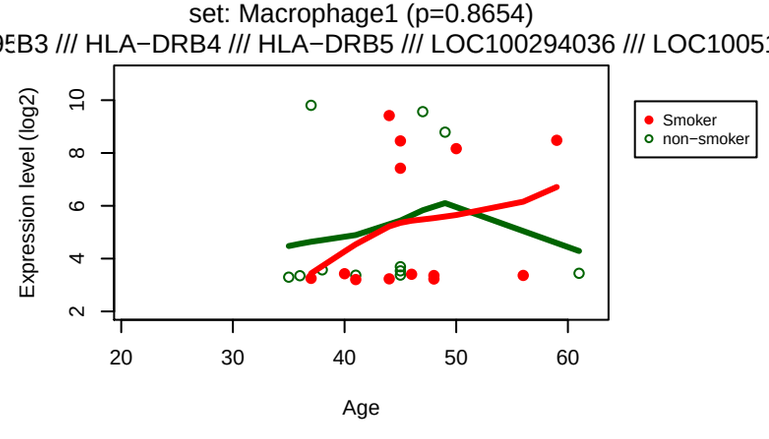

set: SmallAirway1 (p=0.9854)
B3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC10051049
Expression level (log2)
Age
Smoker
non−smoker

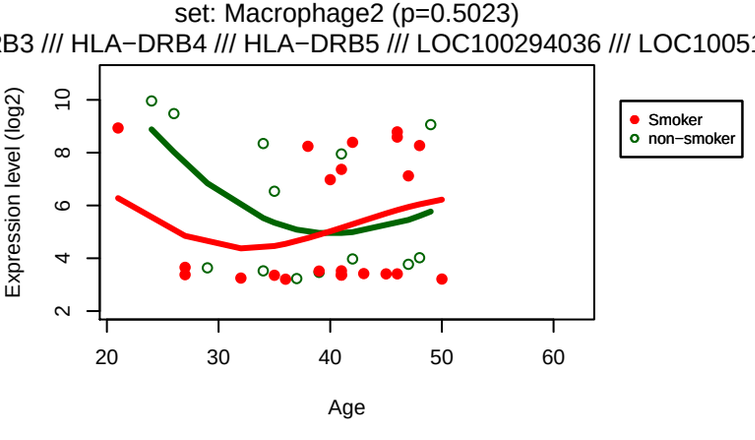

set: SmallAirway2 (p=0.1656)
B3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC10051049
Expression level (log2)
Age
Smoker
non−smoker

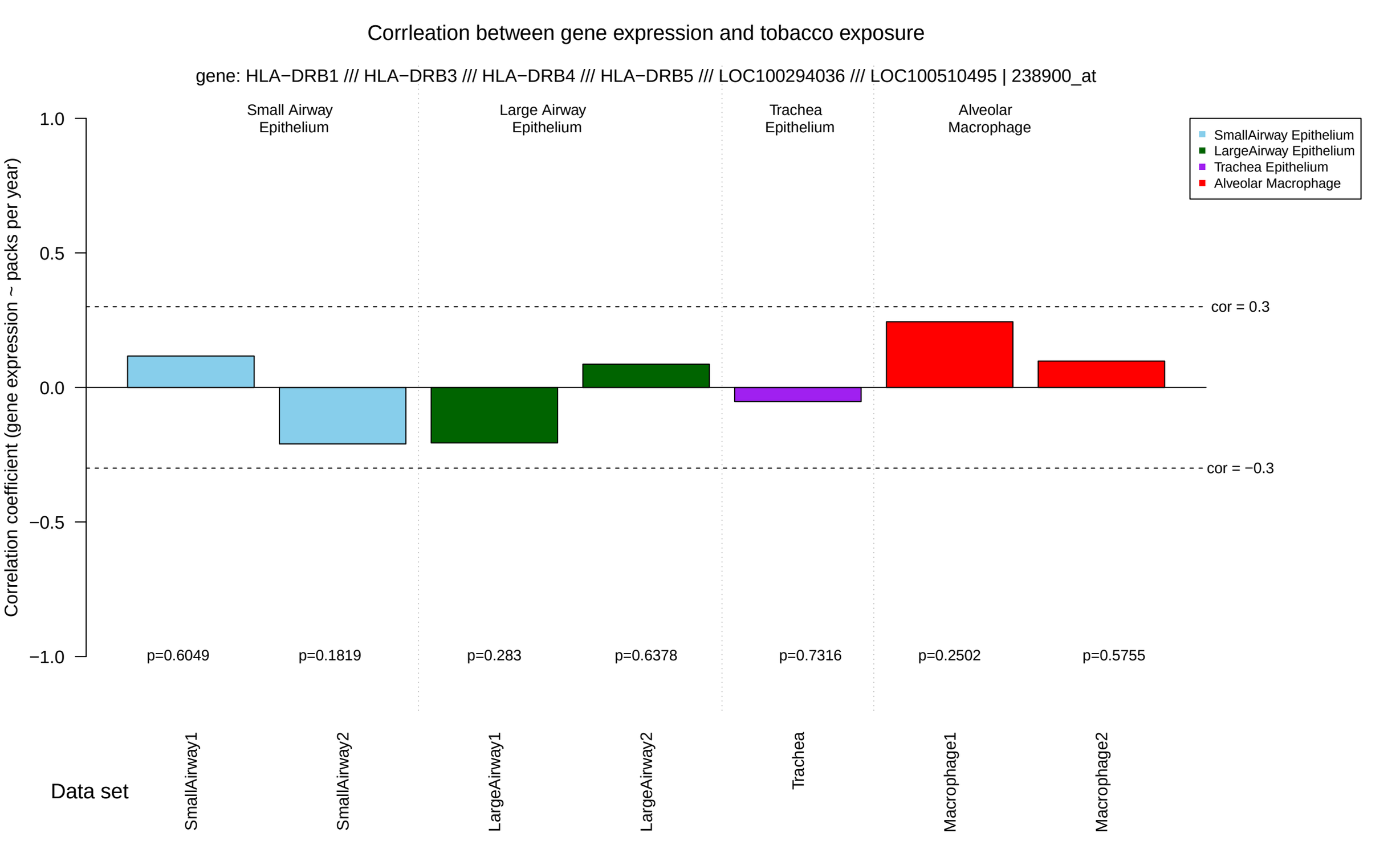

set: LargeAirway1 (p=0.1755)
B3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC100510495
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5943)
B3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC10051049
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3725)
B3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC10051049
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8654)
B3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC100510495
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5023)
B3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC10051049
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 /// HLA−DRB5 /// LOC100294036 /// LOC100510495 | 238900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6049
p=0.1819
p=0.283
p=0.6378
p=0.7316
p=0.2502
p=0.5755
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2